CSA0710
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SLOT - D

27

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I S.Nirmal/192511249 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Nirmal

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	1	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 35 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 **MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-struct ured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization , incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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6/10/2026 CO1 AT1 - Short answers

S.No: 27

⑥ Given Data:

Original file size = 900 MB

Compression rate = 40 %

Encryption overhead = 5 %

To: I. Calculate 1. Compressed file size, 2. encrypted file size, 3. precentage reduction in file size, II. Explain presentation layer.

Calculation:

1. Calculating Compressed file size,

Compressed file size amount = file size × Compression rate

= 900 × 40 % = 900 × 0.4

= 360 MB

Therefore, the final Compressed file

Size = file size - Compressio Amount

= 900 MB - 360 MB

= 540 MB,

2. Calculating encrypted file size,

Encryption overhead = Compression file size
× encryption overhead %.

= 540MB × 5% = 540 × 0.05

= 27MB

Encrypted file size = Compressed file size
+ encryption overhead

= 540 MB + 27MB

= 567MB

3. Calculating total percentage reduction in file size,

Reduction Ratio = Orignal file size - Encrypted file size / original file size

$= \frac{900 - 567}{900} = \frac{333}{900} = 0.37$

File size reduction ratio is

0.37,

Reduction percentage = Reduction Ratio × 100

= 0.37 × 100

∴ Reduction percentage = 37%.

Role of presentation Layer:

⇒ The presentation layer is the Sixth layer of the OSI model, it is responsible for the representation and format of data exchanged between two communicating systems.

⇒ The major functions are:

1. Data Compression: Reduce size of data
2. Data Encryption: protect data.
3. Data translation: Convert data into suitable format.
4. Data formatting: Ensure recieves can understand data.

Conclusion:

→ The original file of 900 MB is Compressed to 540 MB. After adding encryption overhead, the final size becomes 567 MB resulting in a net file size reduction of 37%.

→ roles of presentation layer has also been discussed.

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(9) Given data:

Total file size = 3 GB

Size of each request = 3MB

Effective Throughput = 120 Mbps

To: Calculate total no. of requests time taken & roles of Application Layer

Calculation:

1. Calculation of total no of requests

Taking,

1 GB = 1000 MB

Therefore,

$3\,GB = 3 \times 1000 = 3000\,MB$

$$\text{Number of requests} = \frac{\text{Total file size}}{\text{Request size}}$$

$$= \frac{3000}{3} = 1000 \text{ requests}$$

2. Calculation of transfer time.

We know, 1 byte = 8 bit.

$3\,GB = 3 \times 1000^3$ bytes

$= 24,000,000,000$ bits

Given - Throughput = 120 Mbps $= 120 \times 1000^2$ bits/s

$$\text{Transmisson time } (T) = \frac{\text{Total data in bits}}{\text{Throughput}}$$

$$\Rightarrow T = \frac{24 \times 1000^3}{120 \times 1000^2} = 2 \times 100 = 200 \text{ seconds}$$

Converting to mintues.

Transmission time = 200 seconds

= 3 mintues 20 seconds

3. Role of application layer:

⇒ The application layer is the seventh and topmost layer of the OSI model. It provides an interface between network services and user applications.

⇒ Major functions of application layer include, file transfer, Web services, email services and allowing user access to network services.

Conclusion:

⇒ The application generates 1000 requests

→ Estimated transmission time is 200 seconds (or) 3 mintues & 20 seconds, assuming no protocol overhead and constant throughput

(8) Given data:

Component	Delay
Application Layer	4ms
Transport layer	3ms
Network Layer	5ms
Data link layer	2ms
Physical layer	1ms
Propogation delay	18ms
Transmission delay	12ms

Calculation:

1. Calculating total processing delay.

Total processing delay = A.L + T.L + N.L + DL.L + P.L

$T_P = 4 + 3 + 5 + 2 + 1$

$T_P = 15ms$

Therefore, the total processing delay is 15ms.

2. Calculating total end-to-end delay,

$$T_{Total} = T_P + T_T + T_P$$

$$= 15 + 18 + 12$$

$$= 45ms$$

3. Types of other delays Explanation:

1) Transmission delay: time required to place all the packet bits onto the communication link.

2) propogation delay: time required for the data to move through (propogate) the physical medium.

Conclusion:

⇒ The total processing delay is 15ms

⇒ After adding Transmisson & progation delay the final end-to-end delay is 45ms.

(9) Given data:

→ Useful data = 80 MB

→ Frame Size = 1600 bytes

→ Protocol overhead p/f = 80 bytes.

To: Calculate total no. of frames, total protocol overhead & efficiency.

Calculation:

1. calculating total no. of frames,

Useful data per frame = frame Size − Protocol overhead

= 1600 − 80

= 1520 bytes

We know,

1MB = 1000^2 bytes

⇒ 80MB = 80×1000^2 bytes

Number of frames (N) = Total Useful data / Useful data frame

= $\frac{80 \times 1000^2}{1520}$

$\Rightarrow N = 52,631.58$

$\Rightarrow N = 52632$ frames //

2. Calculating total protocal overhead

Total protocol overhead = $N \times$ Overhead size per frame

$= 52,632 \times 80$

$= 4,210,560$ bytes

$= 4.21$ MB //

3. Calculating transmission efficiency.

$$\text{Transmisson Efficiency (\%)} = \frac{\text{Usefull data}}{\text{Usefull data + Overhead}} \times 100$$

$$= \frac{80\,MB}{80MB + 4.21MB} \times 100 = \frac{80 \times 100}{84.21}$$

$= 95\%$

Therefore, the transmisson efficiency is 95% //

4. Effect of protocol overhead:

⇒ protocol overhead consists of additional info used by the protocol.

⇒ the effects of protocol overhead are:

1. Increased data transmisson
2. Reduced efficency
3. Increased transmisson time
4. Improved communication
5. proper band width utilisation.

Conclusion:

⇒ The network requires 52,632 frames.

⇒ The total protocol overhead is 4.21MB.

⇒ The transmisson efficency is 95%.